

WASP-10b

R-1320

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_211001 · created 2026-07-31T19:29:07+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459488.6967 +/- 0.0047 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.166 +/- 0.031
Transit depth (Rp/Rs)^2	2.74 +/- 1.02 [%]
Orbital Inclination (inc)	89.74 +/- 1.17
Airmass coefficient 1 (a1)	1.05 +/- 0.025
Airmass coefficient 2 (a2)	-0.053 +/- 0.051
Scatter in the residuals of the lightcurve fit is	2.43 %
Best Comparison Star	#1 - [585, 215]
Optimal Aperture	5.42
Optimal Annulus	11.28
Transit Duration (day)	0.0938 +/- 0.0038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.166 ± 0.031 vs 0.1592 (deviation 0.22σ)
Duration fit vs published	0.0938 d vs 0.0946 d (deviation 0.21σ)
Mid-time O–C	4.3 min
Depth SNR	5.4
Residual scatter	2.43 %

Light curve

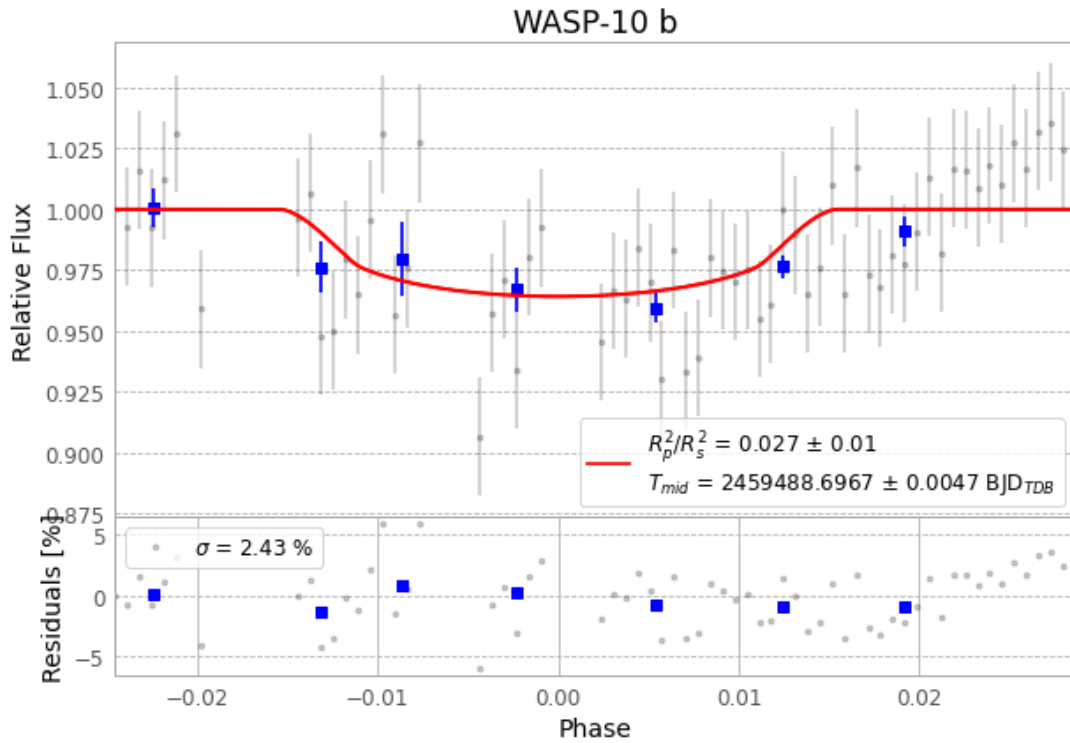


Figure 1 — FinalLightCurve_WASP-10 b_2021-09-30.png (SHA-256 783fd49e1f7e6984...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [357, 204], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f9c51ce660682680...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

80 FITS frames (53 MB), WASP-10211001024512.FITS → WASP-10211001064212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-211001.log (0f8776d433fe...), FinalLightCurve_WASP-10 b_2021-09-30.png (783fd49e1f7e...), FinalLightCurve_WASP-10 b_2021-09-30.csv (57ea1e88581d...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 19:29Z.